SCAR-MLKEM · Internal Architecture

Security-Constrained Adaptive Reconstruction and caching for ML-KEM-768 · per-request decision path on the responder

Online state update after each request (history, arena)

01 · Inputs and frozen constants

Request (j, t): context id, time
Seed (d, z): 64 B per context (FIPS 203)
Ciphertext c: 1088 B

Frozen: C_{ps}, a_P, a_S, e_P, e_S (Pi-calibrated)
$T = 10$ ms, $g = 0.5$ ms, $H = 10$ ms
$k = 0$, $\nu = 2$, margin 0; mask 31

Online state: $(n_j, t_j, \bar{g}_j)$, arena occupancy

02 · Causal forecast (AGE · FORGET)

$$q_j(t) = \frac{n_j}{n_j + 2} \cdot \frac{1}{\max(\bar{g}_j,\ t - t_j,\ 50\,\mu\text{s})}$$

EWMA gap: $\bar{g}_j \leftarrow \frac{1}{2}(\bar{g}_j + \Delta t_j)$
n_j: gaps observed, t_j: last access of j
Bounded history: 1000 contexts (FORGET)

Unnormalised intensity, not a probability

03 · Lifetime-aware value (DEADLINE)

$$R_j = \max\{0,\ \min(H,\ b_j + T - g - t)\}$$
b_j: secret birth, never renewed by hits

$$V_j(p, s) = q_j\,[\,H\,p\,d_P + R_j\,s\,d_S + \rho_j\,p\,s\,d_I\,]$$
$\rho_j = \min(H, R_j)$; $p, s \in \{0, 1\}$ retention flags

$d_P = 9.06$, $d_S = 3.05$, $d_I \approx 0$ *µs (frozen)*

04 · Public admission (COST, ADMISSION)

$$\Delta V_P = V_j(1, s) - V_j(0, s)$$
Charged: $a_P + \mathbf{1}[s = 0]\,d_I + e_P + C_{00}/32$
Victim loss: $\sum_v V_v + i\,C_{00}/1024$, i records

Accept iff $\Delta V_P >$ charged cost + victim loss
First-fit prefix, bounded search; 3584 B/record

Non-positive scores are rejected

05 · Secret admission

$$\Delta V_S = V_j(p, 1) - V_j(p, 0)$$
Accept iff $\Delta V_S > a_S + e_S + C_{00}/32$

Only into space left after the public step
No victims; birth $b_j = t$; core 1216 B

Secrets never displace public records

06 · Lifecycle and clearing

Before use: expiry check $t - b_j \geq T - g$
Expired cores are never borrowed

After use and at the 64-request barrier:
zeroise, verify zeros, record lateness

Late clears are endpoints, not stop rules

07 · Execution by retention state (p, s) · standard ML-KEM-768 Decaps

State (0,0)	**State (1,0)**	**State (0,1)**	**State (1,1)**
expand dk from seed, derive A	borrow public record (A, ek)	borrow secret core	borrow both tiers
$C_{00} = 31.83$ µs	$C_{10} = 22.77$ µs	$C_{01} = 28.78$ µs	$C_{11} = 19.72$ µs

dk: decapsulation key, ek: encapsulation key, A: public matrix
Implicit rejection, re-encryption check and scratch clearing unchanged; algorithm, parameters and key sizes exactly as in FIPS 203

08 · Outputs and telemetry

Shared secret $K \in \{0, 1\}^{256}$
Updated arena and per-context history

Per request: state, hits, admissions, timers
Per process: thread CPU, RSS, late clears

No trainable weights: constants calibrated offline, state updated online